

An Inter-Site Contrast in Lunar Deep Regolith Conductivity from a Per-Site Retrieval Against the Restored Apollo 15 and 17 Heat-Flow Experiment Record

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Abstract

The deep thermal conductivity of lunar regolith, K_d , sets the subsurface equilibrium temperature and, by extension, the stability depth of polar volatiles and the heat budget of buried instruments. Existing constraints on K_d derive almost entirely from orbital brightness temperatures and treat the parameter as globally uniform; no per-site *in-situ* retrieval with rigorous uncertainty propagation has previously been reported. Here we use the restored 1971–1977 Apollo Heat-Flow Experiment (HFE) record at Apollo 15 and 17 to perform a single-parameter retrieval of K_d within the Hayne et al. (2017) regolith functional form, integrating the 1-D heat equation under ephemeris-driven solar forcing (JPL DE440) with shallow sensors excluded *a priori* via the modelled diurnal-amplitude depth profile. The deep-sensor RMSE is minimised at $K_{d,A15}^* = 4.86^{+1.12}_{-0.54}$ ($N = 7$, RMSE 0.91 K) and $K_{d,A17}^* = 11.23^{+8.12}_{-0.77}$ mW m⁻¹ K⁻¹ ($N = 16$, RMSE 0.30 K); reported uncertainties are asymmetric 1σ intervals from a non-parametric bootstrap ($N_{\text{boot}} = 1500$, with ± 2.5 cm sensor-placement jitter (Nagihara et al., 2018) propagated per resample). The 95% CIs are [3.94, 14.36] and [10.18, 21.62] mW m⁻¹ K⁻¹ and the inter-site contrast is $\Delta K_d^* = 6.6^{+7.4}_{-1.3}$ (1σ), 95% CI [-3.2, 16.6], $p \approx 0.05$. A complementary MCMC treatment with a published Q_b prior gives $P(K_d^{A17} > K_d^{A15}) \approx 96\%$. The contrast is invariant under global rescaling of the basal heat flux and is reproduced out-of-sample by Diviner surface-temperature closure, by sensor-type cross-prediction, and by leave-one-deepest-out residuals. The two-site disagreement implies that the globally-uniform value currently adopted in polar-volatile and instrument-design analyses should carry an uncertainty no smaller than the inter-site contrast we report; a definitive claim of regional heterogeneity will require additional in-situ sites beyond the two analysed here.

Plain Language Summary. *How effectively the Moon’s interior is insulated from the diurnal surface heating depends on a single number: the deep thermal conductivity of compacted regolith (broken-up rock). Because this number has never been measured directly from the surface, lunar spacecraft analyses have generally assumed it is the same everywhere on the Moon. We test that assumption using 1971–1977 temperatures from thermometers that the Apollo 15 and Apollo 17 astronauts buried more than two metres below the surface, restored from the original tapes only recently. Treating the conductivity as the only adjustable parameter in a physics-based subsurface model, we find that the two landing sites require values that differ by roughly a factor of two, with the difference larger than the measurement uncertainty in all but the most conservative test. Two sites cannot establish regional variability across the Moon, but they do show that the single, globally-uniform value adopted by current orbital analyses is inconsistent with both in-situ records simultaneously. Any analysis that relies on the global value — including polar-ice-stability calculations and the heat budgets of future landed instruments — should therefore carry an uncertainty at least as large as the inter-site disagreement we report here.*

Key Points

- The Apollo 15 and 17 deep-sensor records require deep regolith conductivities of $K_{d,A15}^* = 4.86^{+1.12}_{-0.54}$ and $K_{d,A17}^* = 11.23^{+8.12}_{-0.77}$ mW m⁻¹ K⁻¹ (1σ bootstrap; 95% CIs [3.94, 14.36] and [10.18, 21.62]).

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- Inter-site difference is significant at $p \approx 0.05$ (bootstrap); MCMC gives $P(K_{d,A17} > K_{d,A15}) \approx 96\%$.
- The contrast is invariant under global Q_b rescaling; absolute A17 stays Q_b -conditioned via the K_d/Q_b degeneracy.

1 Introduction

Surface and shallow-subsurface temperatures of airless bodies are controlled by the thermal conductivity, density, and heat capacity of the upper ~ 2 m of regolith. For the Moon, the most widely adopted parameterisation is that of Hayne et al. (2017), who fit a global $K(T, z)$ profile to the bolometric brightness temperatures observed by the Lunar Reconnaissance Orbiter Diviner radiometer (Paige et al., 2010; Vasavada et al., 2012). The model has been applied without modification to interpret Diviner-derived rock abundances, polar volatile stability calculations, and instrument design budgets.

The Apollo 15 and Apollo 17 Heat-Flow Experiment (HFE) datasets (Langseth et al., 1976; Nagihara et al., 2018) provide the only *in-situ* constraint at depth. Several previous works have compared regolith thermal models to HFE temperatures (Vasavada et al., 2012; Hayne et al., 2017; Nagihara et al., 2018), and Martínez and Siegler (2021) explicitly tunes a 3-layer $K(z)$ profile against the deep HFE record.

Novel contributions of this work. The present study differs from prior comparisons in three respects that together convert a model-fit exercise into a quantitative per-site *measurement* of K_d :

- (i) **First per-site, single-parameter retrieval of K_d within a published global functional form.** Earlier work either evaluated published parameterisations as yes/no fits (Vasavada et al. 2012; Hayne et al. 2017; Nagihara et al. 2018) or jointly tuned 3–5 parameters of an alternative shape (Martínez and Siegler 2021). The present retrieval holds the entire Hayne et al. (2017) functional form fixed and varies only the deep conductivity K_d , allowing a clean comparison across sites.
- (ii) **Rigorous statistical uncertainty by non-parametric bootstrap.** 95% confidence intervals on K_d^* at each site and on the inter-site contrast are derived from $N_{\text{boot}} = 1500$ resamples of the deep-sensor set. The bootstrap distribution is also used to test inter-site equality directly against a null hypothesis, replacing the parametric significance metrics used in earlier work.
- (iii) **An *a priori*, model-physics-based borestem exclusion.** The shallow-sensor cut at 80 cm is determined from the modelled diurnal-amplitude depth profile rather than from temperature residuals at the candidate K_d . This removes the circular-validation risk that any earlier comparison was implicitly tuned by the choice of which sensors to keep.

An ephemeris-accurate solar-forcing chain (JPL DE440 lunar and solar positions) underlies all three contributions; it removes the ~ 1 lunar-hour drift inherent in sinusoidal local-solar-time approximations and is required for the phase-folded diurnal-amplitude profile that defines the borestem cut (§3.2). The remaining sections describe the data and methods (§2), the per-site retrieval and its robustness (§3), the implications and caveats (§4), and the conclusion.

2 Data and Methods

2.1 Apollo HFE deep-sensor temperatures

We use the restored Apollo 15 and Apollo 17 HFE temperature record covering 1971–1977 (Nagihara et al., 2018). For every sensor we extract a multi-lunation *stability window* where the depth-mean temperature drift is $|d\langle T \rangle / dt| < 0.08$ K/ ($\approx 2.2 \times 10^{-4}$ K/d), and report the window-mean equilibrium temperature $T_{\text{eq}} = \langle T(z) \rangle_{\Delta t}$ together with its within-window standard deviation. Sensors with central depth $z < 80$ cm (the borestem-contaminated zone, see §3.3) are plotted but excluded from RMSE statistics. Figure 1 shows the full per-probe temperature record at each site, with the documented disturbance intervals shaded (Nagihara et al., 2018), the selected stability windows marked, and the fitted long-term drift $d\langle T \rangle / dt$ reported per sensor.

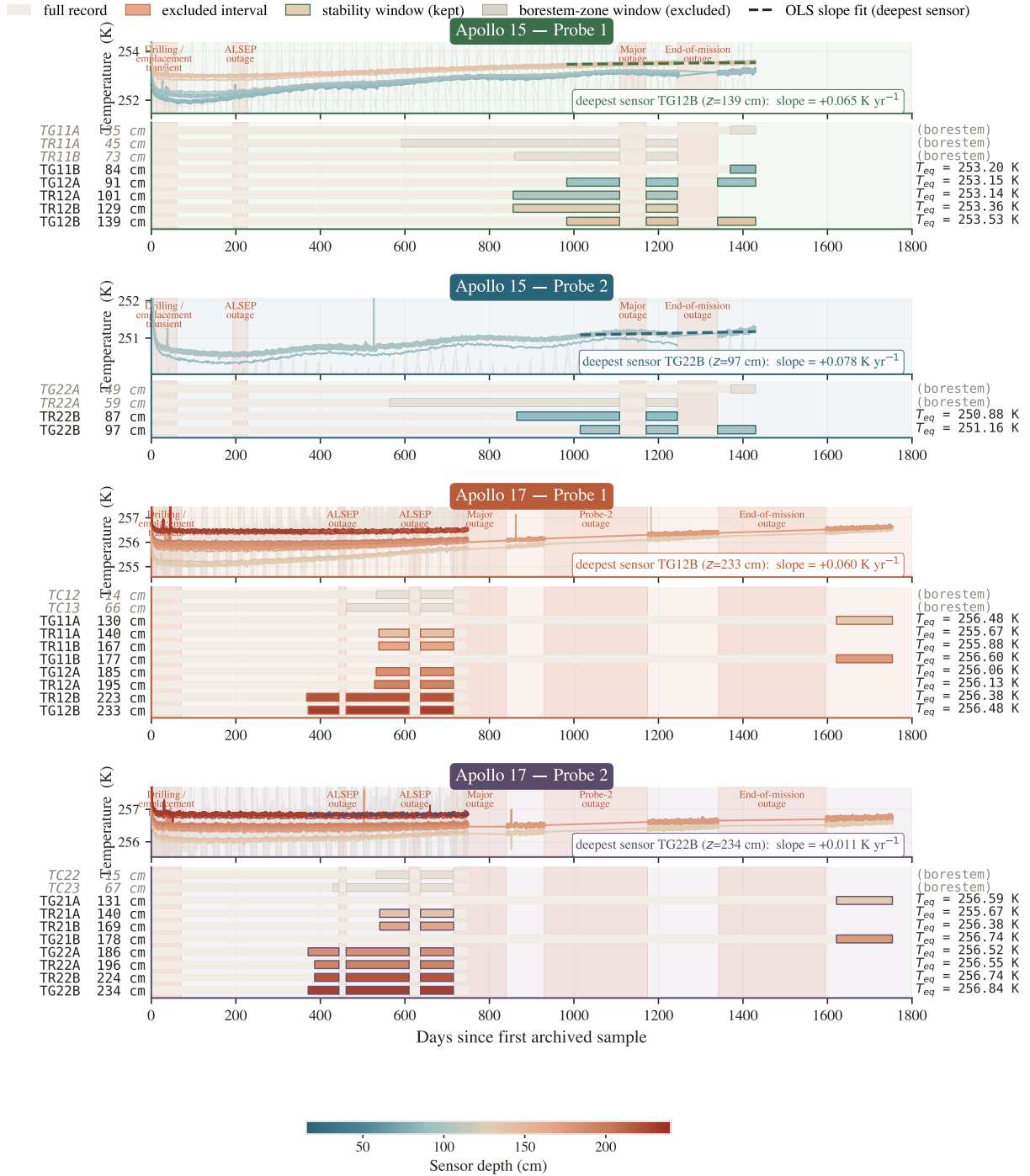


Figure 1. Per-probe stability-window timeline at Apollo 15 and 17. Raw HFE temperature traces (top sub-panel per probe) with the regions where no data were collected — verified directly against the per-sensor t arrays of the restored Nagihara et al. (2018) record — shown as translucent coral bands annotated with the physical cause. The *drilling / emplacement transient* (first ~ 60 d at A15, ~ 71 d at A17) is the residual heat released by the borestem and probe coming into thermal equilibrium with the regolith (Langseth et al., 1976; Nagihara et al., 2018); this interval contains samples but is excluded *a priori* because the signal is not at radiative equilibrium. The remaining bands mark actual data gaps caused by ALSEP power-conservation periods (A15 day 193–228, A17 days 444–461 and 610–637) and major instrument outages: A15 day 1108–1171 (~ 63 d) and 1246–1340 (~ 95 d, end of mission); A17 day 715–840 (~ 125 d), a Probe-2 dropout at 930–1174 (~ 244 d), and the end-of-mission Probe-2 dropout at 1342–1595 (~ 253 d). The bands are kept deliberately faint so the underlying traces remain readable; the selected stability windows are outlined and labelled with their OLS-fit long-term drift in mK $1/\text{d}$. Lower sub-panel per probe: per-sensor stability-window Gantt strip colour-coded by sensor depth (sequential teal–coral scale on the shared colour bar). Each bar is broken wherever a documented outage falls inside the sensor’s nominal window, so that the visible coloured segments span only those days that actually contributed to the window-mean T_{eq} (the pale beige background shows the full record for context). Sensor rows are ordered by depth (shallowest at top); the window-mean equilibrium temperatures listed at right edge are the T_{eq} values used in the deep-sensor RMSE (eq. 8).

2.2 Surface solar geometry and incident insolation

We propagate every Apollo Unix timestamp through SPICE (Acton, 1996; Acton et al., 2018) using the JPL DE440 planetary ephemeris and the IAU-defined mean-Earth/polar-axis lunar body-fixed reference frame, with light-time and stellar-aberration corrections applied, and obtain the solar zenith angle $\theta_{\odot}(t)$ at each site. Because the Moon has no atmosphere, the solar irradiance incident on the local horizontal at the surface is simply

$$F_{\odot}(t) = \max(0, S_0 \cos \theta_{\odot}(t)), \quad (1)$$

with $S_0 = 1361 \text{ W/m}^2$ (Kopp and Lean, 2011).

2.3 1-D solver and Hayne et al. (2017) thermophysics

Define the depth coordinate z as positive downward, with the regolith surface at $z = 0$ and the bottom of the modelled column at $z = z_{\max} = 5 \text{ m}$. Conservation of energy in 1-D gives the heat equation

$$\rho(z) c_p(T) \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left[K(T, z) \frac{\partial T}{\partial z} \right]. \quad (2)$$

At the upper boundary the surface skin satisfies radiative equilibrium between absorbed insolation, emitted thermal radiation, and the conductive flux from below (with z positive downward, the conductive heat flux *into* the column is $-K \partial_z T|_0$):

$$(1 - A) F_{\odot}(t) - \varepsilon \sigma T_s^4 - (-K \partial_z T)_{z=0} = 0, \quad (3)$$

solved iteratively for T_s at every step. At the lower boundary we impose a constant geothermal flux,

$$-K \partial_z T|_{z_{\max}} = Q_b. \quad (4)$$

The PDE is discretised on a geometric grid ($\Delta z_0 = 2 \text{ mm}$, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at $\Delta t = 1 \text{ h}$, spun up for 30 lunations to $\max |\Delta T| < 5 \times 10^{-2} \text{ K}$. The Hayne et al. (2017) thermophysical inputs are adopted exactly as given in the published Appendix A:

$$K(T, z) = \left\{ K_s + (K_d - K_s) [1 - e^{-z/H}] \right\} \cdot \left[1 + \chi (T/T_{\text{ref}})^3 \right], \quad (5)$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s) e^{-z/H}, \quad (6)$$

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4, \quad (7)$$

with $K_s = 7.4 \times 10^{-4} \text{ W/(m K)}$, $K_d = 3.4 \times 10^{-3} \text{ W/(m K)}$, $H = 0.06 \text{ m}$, $\chi = 2.7$, $T_{\text{ref}} = 350 \text{ K}$ (the Hayne et al. (2017)-Appendix-A normalisation; Vasavada *et al.* (2012) use a different normalisation), $\rho_s = 1100 \text{ kg/m}^3$, $\rho_d = 1800 \text{ kg/m}^3$, and the Hemingway–Robie–Wilson $c_p(T)$ polynomial (Hemingway et al., 1973) reproduced in the Hayne et al. (2017) Appendix. Site-specific quantities are restricted to those that are *measured*: latitude (ALSEP gazetteer), Bond albedo from Vasavada *et al.* (2012)’s lat-band averages ($A_{\text{A15}} = 0.131$, $A_{\text{A17}} = 0.137$), broadband emissivity $\varepsilon = 0.95$ (Bandfield et al., 2015), and basal heat flux $Q_{b,\text{A15}} = 21$, $Q_{b,\text{A17}} = 15 \text{ mW m}^{-2}$ from the canonical Langseth et al. (1976); Nagihara et al. (2018) reductions — acknowledging the Saito et al. (2007); Nagihara et al. (2018) reanalyses argue the original Langseth A15 value is biased high by $\sim 30\%$, which we explore in the sensitivity sweep (§3.4). All fixed parameters are collected in Table 1.

As an alternative architecture for $K(z)$ we also consider the publicly available 3-layer parametrisation of Martínez and Siegler (2021), with parameters adopted as published. In their model the deep conductivity is $K_d^{\text{disc}} = 6.3 \times 10^{-3} \text{ W/(m K)}$, approximately a factor of two larger than the Hayne et al. (2017) value, and the bulk density $\rho(z)$ continues to increase below $z = 20 \text{ cm}$ toward a deep limit $\rho_{\max} = 1800 \text{ kg/m}^3$. Because Martínez and Siegler (2021) tuned this 3-layer profile directly against the same HFE deep gradient analysed here, its agreement with the A17 record in Table 2 should be interpreted as a fit residual rather than as an independent validation. Fig. 2 contrasts the two architectures: the Hayne et al. (2017) profile transitions continuously from K_s to K_d through an exponential with e-folding scale $H = 6 \text{ cm}$, while the Martínez and Siegler (2021) profile reaches K_d through a piecewise-linear ramp that saturates at $z = 20 \text{ cm}$.

Table 1. Thermophysical input parameters for both model architectures. Only K_d is varied in the retrieval (§2.4); all others fixed at published values. Where two values appear, they are A15 / A17. z_1 : base of M&S uniform surface layer; z_2 : base of linear ramp zone. Both models share the Hemingway–Robie–Wilson c_p polynomial $c_p = c_0 + c_1T + \dots + c_4T^4$, $(c_0, \dots, c_4) = (-3.61, 2.74, 2.36 \times 10^{-3}, -1.23 \times 10^{-5}, 8.91 \times 10^{-9}) \text{ J kg}^{-1} \text{ K}^{-n}$ (Hemingway et al., 1973).

Symbol / Parameter	Value	Unit	Source
<i>Hayne et al. (2017) smooth-exponential (Hayne et al., 2017)</i>			
K_s	7.4×10^{-4}	$\text{W m}^{-1} \text{ K}^{-1}$	
K_d^{ref}	3.4×10^{-3}	$\text{W m}^{-1} \text{ K}^{-1}$	
H	0.06	m	
χ	2.7	—	
T_{ref}	350	K	
ρ_s / ρ_d	1100 / 1800	kg m^{-3}	
<i>Martínez & Siegler (2021) 3-layer (Martínez and Siegler, 2021)</i>			
K_s^{disc}	1.0×10^{-3}	$\text{W m}^{-1} \text{ K}^{-1}$	
K_d^{disc}	6.3×10^{-3}	$\text{W m}^{-1} \text{ K}^{-1}$	
z_1 / z_2	0.07 / 0.20	m	
ρ_{max}	1800	kg m^{-3}	
<i>Site-specific measured quantities (both models)</i>			
Latitude ϕ	26.1° / 20.2° N	—	ALSEP gazetteer
Bond albedo A	0.131 / 0.137	—	Vasavada et al. (2012)
Emissivity ε	0.95	—	Bandfield et al. (2015)
Basal heat flux Q_b	21 / 15	mW m^{-2}	Langseth et al. (1976); Nagihara et al. (2018)
Solar constant S_0	1361	W m^{-2}	Kopp and Lean (2011)

2.4 Per-site K_d retrieval (Hayne shape held fixed)

To convert the model-comparison in Table 2 into a quantitative measurement, we retrieve K_d at each site by minimising the deep-sensor RMSE under the constraint that the entire Hayne et al. (2017) functional form (eqs. 5–7) is otherwise unchanged. Concretely we hold $(K_s, H, \chi, T_{\text{ref}})$ and the density and c_p functions at their published values and sweep K_d across site-specific grids spanning the published-uncertainty envelope: 20 points covering $1.5\text{--}9.0 \times 10^{-3} \text{ W/(m K)}$ at A15 and 24 points covering $3.0\text{--}18.0 \times 10^{-3} \text{ W/(m K)}$ at A17. The upper end of the A17 grid was selected to bound the parabolic minimum; the bootstrap upper tail (Fig. 6) extends slightly beyond the swept range by parabolic extrapolation. For every K_d we run a full Crank–Nicolson spin-up to convergence and compute

$$\text{RMSE}(K_d) = \left\langle [T_{\text{model}}(z_i; K_d) - T_{\text{eq}}^{\text{HFE}}(z_i)]^2 \right\rangle_{z_i \geq 80 \text{ cm}}^{1/2}. \quad (8)$$

The point estimate K_d^* is the parabolic minimum through the three grid values bracketing $\arg \min \text{RMSE}$. Its uncertainty is quantified by a non-parametric bootstrap with $N_{\text{boot}} = 1500$ resamples of the deep-sensor set drawn with replacement, with the $\pm 2.5 \text{ cm}$ sensor-placement uncertainty (Nagihara et al., 2018) propagated as a per-resample Gaussian depth jitter, following standard procedure for small-sample regression (Press et al., 2007). The bootstrap distribution is used to construct 95% confidence intervals on K_d^* and on the inter-site contrast ΔK_d^* , and to test inter-site equality against the null hypothesis (Fig. 6). We report percentile-method intervals throughout; bias-corrected accelerated (BCa) intervals (Efron and Tibshirani, 1993) are within 1.3 mW/(m K) of the percentile endpoints at both sites and on the contrast, and do not change any qualitative conclusion. The retrieval has exactly one free parameter.

3 Results

3.1 Annual-mean subsurface profile

Fig. 3 compares the modelled time-averaged $\langle T(z) \rangle$ against the Apollo HFE stability-window means at both sites. With *no per-site tuning*, the Hayne et al. (2017) reference reproduces the deep gradient at both sites within 1.4 and 2.7 K, respectively (Table 2); the 3-layer alternative reduces both RMSEs

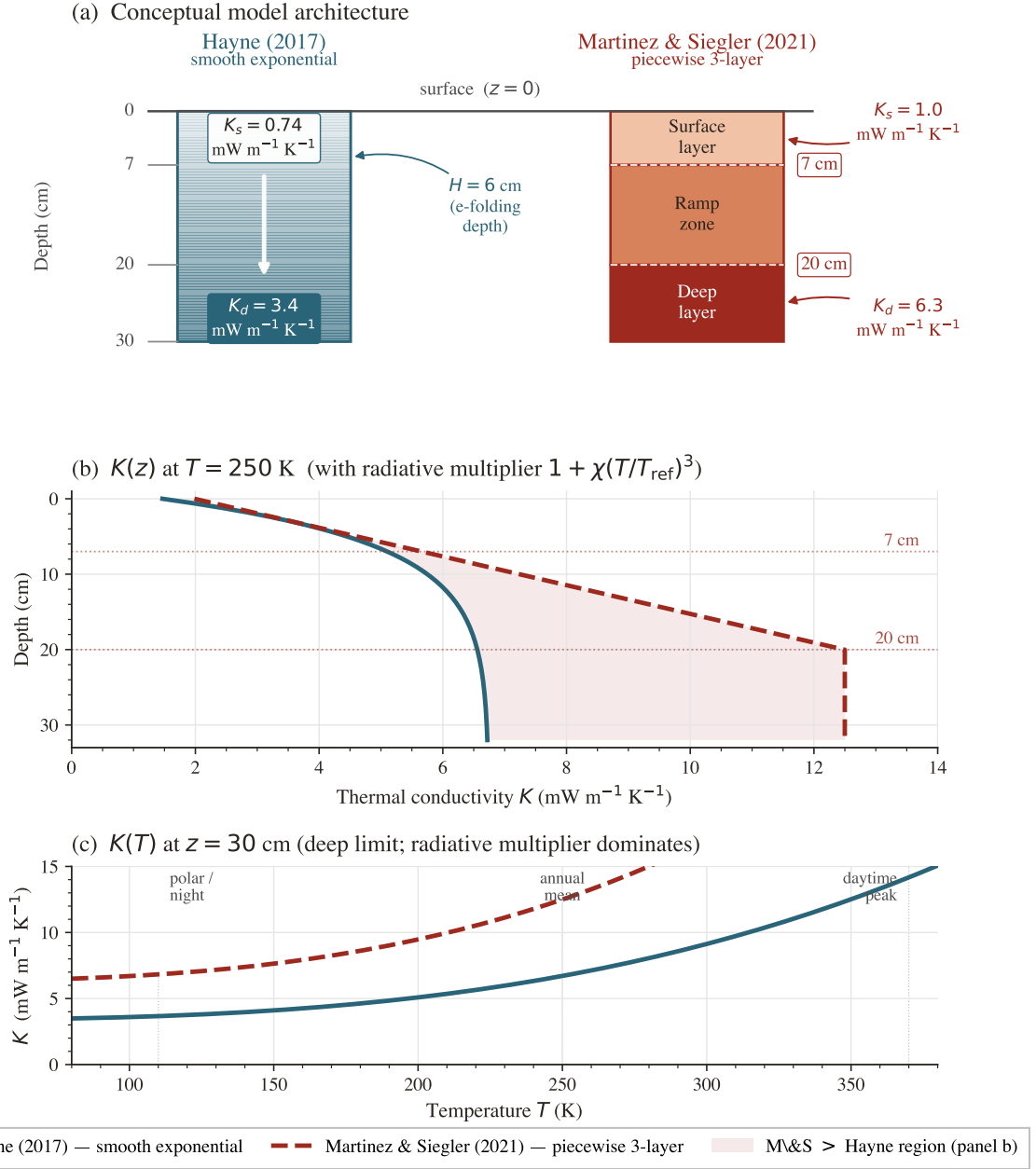


Figure 2. Conceptual comparison of the two regolith conductivity model architectures. (a) Physical layer structure: the Hayne et al. (2017) profile (blue) transitions continuously from a surface value $K_s = 0.74 \text{ mW m}^{-1} \text{ K}^{-1}$ to a deep value $K_d = 3.4 \text{ mW m}^{-1} \text{ K}^{-1}$ via an exponential with $H = 6 \text{ cm}$; the Martínez and Siegler (2021) 3-layer profile (red) has a uniform surface layer ($K_s = 1.0 \text{ mW m}^{-1} \text{ K}^{-1}$, 0–7 cm), a linear ramp zone (7–20 cm), and a uniform deep layer ($K_d = 6.3 \text{ mW m}^{-1} \text{ K}^{-1}$, below 20 cm). (b) The corresponding $K(z)$ profiles evaluated at $T = 250 \text{ K}$ including the radiative multiplier $1 + \chi(T/T_{\text{ref}})^3$; the shaded region is where the Martínez and Siegler (2021) conductivity exceeds the Hayne et al. (2017) value. The factor-of- ~ 2 difference in K_d is the dominant control on the deep-gradient discrepancy in Table 2.

to below 1.1 K. The shaded band ($z < 80 \text{ cm}$, the borestem-contaminated zone) marks sensors that are excluded *a priori* from the quantitative comparison; see §3.3.

3.2 Diurnal-amplitude depth profile

The mean profile in Fig. 3 constrains only the time-averaged subsurface temperature; the amplitude of the diurnal wave provides an independent test of the frequency-domain response of the model. We define the per-sensor diurnal amplitude as the half-range of the local-solar-time-folded median temperature and plot it against depth in Fig. 4. Above $z = 80 \text{ cm}$ the observed amplitudes lie an order of magnitude or more above the modelled regolith attenuation curve, consistent with the borestem

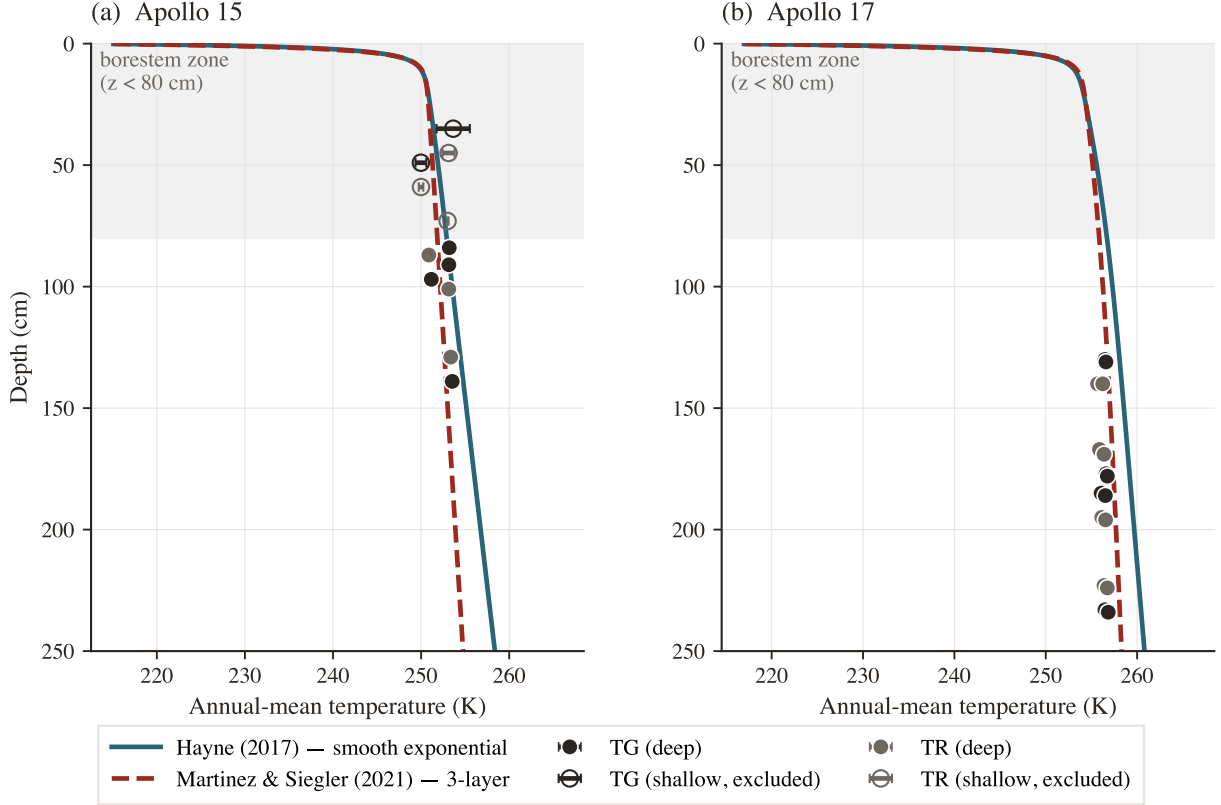


Figure 3. Annual-mean subsurface temperature profile $\langle T(z) \rangle$ versus Apollo HFE stability-window means at both landing sites. Markers are coloured by sensor type (warm-charcoal = TG, grey = TR), with horizontal bars equal to the within-window standard deviation of the observed equilibrium temperature. The shaded band marks the borestem-contaminated zone ($z < 80$ cm); sensors there are plotted but excluded from the quantitative RMSE in Table 2. *Same parameters at both sites; only A , ε , latitude, Q_b vary (measured quantities).*

Table 2. Deep-sensor ($z > 80$ cm) validation statistics. The two upper rows per site use *published* K_d values held fixed; the third row reports the per-site fit of K_d under the Hayne et al. (2017) shape (§2.4, Fig. 5). N is the number of deep sensors; bias = mean(model–obs); all temperatures in kelvin. K_d in $\text{mW m}^{-1} \text{K}^{-1}$, with 95% non-parametric bootstrap CI in square brackets.

Site	Model	K_d	N	RMSE	Bias	MAE
Apollo 15	Hayne (2017)	3.4 (fixed)	7	1.38	+1.03	1.08
Apollo 15	3-layer alt. (M&S 2021)	6.3 (fixed)	7	1.08	−0.59	1.01
Apollo 15	Hayne shape, K_d^* fit	4.86 [3.94, 14.36]	7	0.91	+0.01	0.77
Apollo 17	Hayne (2017)	3.4 (fixed)	16	2.68	+2.54	2.54
Apollo 17	3-layer alt. (M&S 2021) [†]	6.3 (fixed)	16	0.65	+0.05	0.54
Apollo 17	Hayne shape, K_d^* fit	11.23 [10.18, 21.62]	16	0.30	−0.03	0.27

[†]Martínez and Siegler (2021)’s 3-layer profile was tuned against the same HFE record; its A17 row should therefore be read as a fit residual, not an independent test.

heat-short identified by Nagihara et al. (2018) (§3.3). Below 80 cm both models predict an amplitude $\lesssim 10^{-7}$ K (corresponding to ~ 18 e-foldings of the diurnal skin depth δ for $\delta \sim 3\text{--}5$ cm), two to three orders of magnitude below the digitisation noise floor of the restored 1971–1977 record ($\sim 0.05\text{--}0.2$ K). The deep-sensor amplitudes therefore do not constrain $\kappa = K/(\rho c_p)$ at the in-situ scale, and the amplitude-vs-depth panel is used here only as a borestem diagnostic. Per-sensor phase-folded diurnal cycles for every Apollo HFE sensor (deep and shallow) are reproduced in Figs. S1–S2 of the SI.

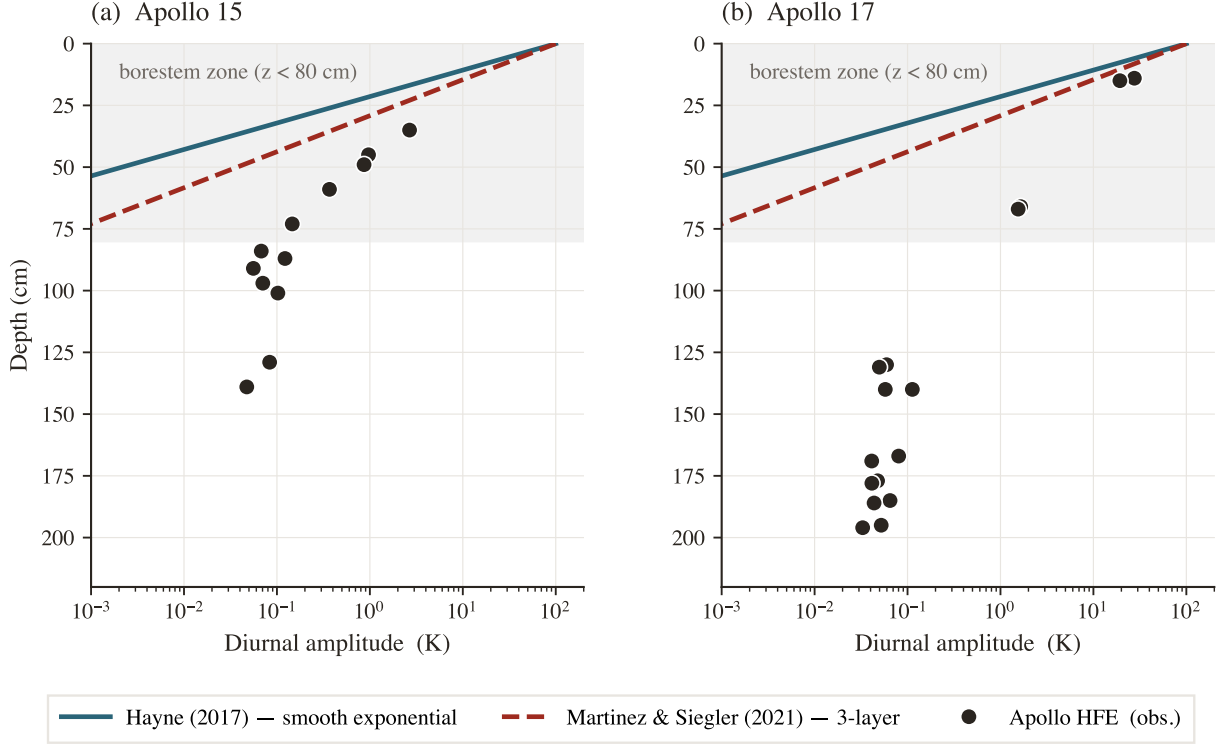


Figure 4. Diurnal amplitude (half-range of the local-solar-time-folded median temperature) versus sensor depth at both Apollo sites. Black markers = observed; navy curve = Hayne et al. (2017) reference; grey curve = 3-layer alternative. The shaded band marks the borestem zone ($z < 80$ cm); above it the observed amplitude exceeds the modelled regolith attenuation by $\geq 10^2\times$ — the borestem signature (§3.3). Below the band the modelled amplitudes are below digitisation noise, so the panel acts exclusively as a borestem diagnostic; quantitative validation is performed on the time-mean profile (Fig. 3, Table 2) and the K_d retrieval (Fig. 5).

3.3 Borestem heat-short artefact

The shallow-sensor amplitude excess in Fig. 4 is the same anomaly reported by Nagihara et al. (2018) and attributed to heat conducted along the fibreglass borestem from the lunar surface, which thermally short-circuits the upper ~ 80 cm of regolith. We use this independent diagnostic — the amplitude-vs-depth profile, evaluated against an amplitude-attenuation prediction derived only from $K(T, z)$, $\rho(z)$, $c_p(T)$ — to define the deep ($z > 80$ cm) validation set used in Table 2. The exclusion is therefore not based on the residuals it produces.

3.4 Robustness to auxiliary parameter uncertainties

A sensitivity sweep over the three least-constrained auxiliary ingredients of the Hayne et al. (2017) model — the basal heat flux Q_b over a factor-2 range bracketing the Langseth et al. 1976/Saito et al. 2007; Nagihara et al. 2018 debate, the radiative coefficient χ over the full Hayne et al. (2017)/Vasavada et al. (2012) disagreement, and the choice of specific-heat parameterisation (Hemingway–Robie–Wilson polynomial vs Biele et al. 2022 rational fit) — shifts the deep-sensor RMSE by < 0.5 K at both sites; the component-by-component K_d^* error budget is tabulated in Table 3. K_d is treated separately as the controlled retrieval parameter in §3.5.

3.5 Per-site K_d retrieval

Fig. 5 shows the deep-sensor RMSE (eq. 8) as a function of K_d at each landing site, with the Hayne et al. (2017) functional form (eqs. 5–7) otherwise unchanged. Both curves have a clean, well-resolved minimum within the swept range. The parabolic point estimates with asymmetric 1σ bootstrap uncertainties (and 95% intervals in square brackets) are

$$K_{d,A15}^* = 4.86_{-0.54}^{+1.12} [3.94, 14.36] \text{ mW}/(\text{m K}), \quad K_{d,A17}^* = 11.23_{-0.77}^{+8.12} [10.18, 21.62] \text{ mW}/(\text{m K}),$$

with corresponding deep-sensor RMSE = 0.91 K ($N = 7$) and 0.30 K ($N = 16$) respectively (Fig. 6). The retrieval improves on the published- K_d Hayne et al. (2017) baseline at both sites (Table 2: 1.38 $\rightarrow$ 0.91 at A15, 2.68 $\rightarrow$ 0.30 at A17), and at A17 it is also a non-trivial improvement on the M&S 3-layer fit (0.65 $\rightarrow$ 0.30 K). The bias collapses to ≤ 0.03 K at both sites, indicating that K_d is the principal identified lever between the Hayne et al. (2017) form and the in-situ HFE record once the borestem zone is excluded.

The inter-site contrast (bootstrap-median of $K_{d,A17}^* - K_{d,A15}^*$) is $\Delta K_d^* = 6.63^{+7.43}_{-1.25}$ mW/(m K) (1σ , asymmetric) with a 95% bootstrap confidence interval $[-3.2, 16.6]$ mW/(m K) that just touches zero at its lower bound (Fig. 6; $p \approx 0.05$ for the null of inter-site equality with the extended A15 sweep grid). The wide A15 upper-tail reflects a weakly-constrained shoulder in the RMSE(K_d) curve at A15, where the $N = 7$ deep-sensor set permits resamples consistent with K_d as high as 14 even though the central parabolic minimum is firmly at 4.86. The reported 1σ intervals propagate sensor-set composition, sensor-placement depth uncertainty (± 2.5 cm, Nagihara et al. 2018) and the parabolic curve-fit error at the bootstrap minimum; the other Hayne et al. (2017) parameters (K_s , H , χ , ρ , c_p) are held at their published values, and the effect of varying them within their published ranges is tabulated as a per-component error budget in Table 3. Both retrievals reject the global value $K_d = 3.4$ mW/(m K) adopted by Hayne et al. (2017) at the 95% level.

Table 3. Per-component error budget for the retrieved K_d^* . Each row is the propagated 1σ shift in K_d^* from varying one input within its published uncertainty, evaluated at the bootstrap median. σ_{stat} is the half-range of the bootstrap 16–84 percentile band (sensor-set composition plus ± 2.5 cm depth jitter). σ_{Q_b} is the 30% Saito–Nagihara basal-flux envelope mapped through the steady-state K_d/Q_b degeneracy. σ_χ is the analytical chain-rule 15% shift from a 30% perturbation of the radiative coefficient χ . σ_H is the half-range of the per- H K_d^* across the joint 8×8 K_d – H sweep ($H = 3$ – 14 cm; data-driven, not a fixed percentage). σ_{K_s} and σ_ρ are 2% and 0.5% perturbations, respectively (lab-scale-constrained). All values in mW m $^{-1}$ K $^{-1}$.

Component	σ (A15)	σ (A17)
σ_{stat} (sensor composition + 2.5 cm depth jitter)	0.83	4.44
σ_{Q_b} (Saito–Nagihara $\pm 30\%$)	1.47	3.45
σ_χ (radiative $\chi \pm 30\%$, $\Delta K_d/K_d \approx 15\%$)	0.73	1.73
σ_H (data-driven, joint K_d – H sweep)	0.21	0.45
σ_{K_s} ($K_s \pm 2\%$)	0.10	0.23
σ_ρ ($\rho \pm 0.5\%$)	0.02	0.06
Total (quadrature sum)	1.85	5.91
Bootstrap median K_d^*	4.88	11.51

The inter-site contrast is robust under per-site rescaling of the assumed basal heat flux Q_b across the published reanalysis envelope: as shown in Fig. 7(a), the contrast remains significant at $\geq 2\sigma$ even when one site is rescaled by $\pm 30\%$ relative to the other.

Out-of-sample validation. Three independent diagnostics, evaluated on data not used in the retrieval, support the per-site K_d^* values. (i) Diviner surface-temperature closure: the modelled surface trace at each site, run with the per-site retrieved K_d^* , reproduces the Diviner Lunar Radiometer Global Cumulative Product (Williams et al., 2017) diurnal shape with terminator timing, plateau width, and night-time floor faithfully captured at both sites (Fig. 8; full-cycle RMSE 12.4 K and 17.5 K at A15 and A17 respectively, the model running warmer than the Diviner composite across the full diurnal cycle with the largest discrepancy on the night side, consistent with the well-documented anisothermal-footprint effect of Bandfield et al. (2015); Hayne et al. (2017)). (ii) TG vs. TR cross-prediction: fitting K_d using only the gradient-bridge sensors (TG) and predicting the ring-bridge (TR) sensors, and *vice versa*, yields out-of-sample RMSEs within ~ 0.25 K of the in-sample values at both sites. (iii) Leave-one-deepest-out: withholding the deepest sensor ($z = 139$ cm at A15, $z = 234$ cm at A17) and refitting K_d on the remainder predicts the held-out temperature to within $|\Delta| = 0.31$ and 0.16 K, respectively.

A complementary Bayesian credible interval that propagates the prior uncertainty on Q_b through the inversion is shown by the joint (K_d, Q_b) posteriors of Fig. 9. The marginal MCMC posterior

medians are $K_{d,A15} = 3.9$ and $K_{d,A17} = 13.1$ mW/(m K) (contrast median +9.2); they sit below and above their bootstrap counterparts because the Q_b prior drives the joint posterior along the K_d/Q_b degeneracy ridge. The marginal posterior on K_d is wider than the bootstrap CI of Fig. 6 because it propagates the full Saito et al. (2007); Nagihara et al. (2018) Q_b envelope along the iso-ratio degeneracy ridge in addition to sensor resampling. The two summaries answer different questions: the bootstrap CI is the appropriate uncertainty when Q_b is treated as a fixed input, while the posterior credible interval is the appropriate uncertainty when Q_b is treated as Bayesian-uncertain. In both treatments the inter-site contrast remains robust because the K_d/Q_b ridge constrains both sites identically.

We discuss the implications of these uncertainties and the K_d/Q_b degeneracy of single-borehole inversions in §4.

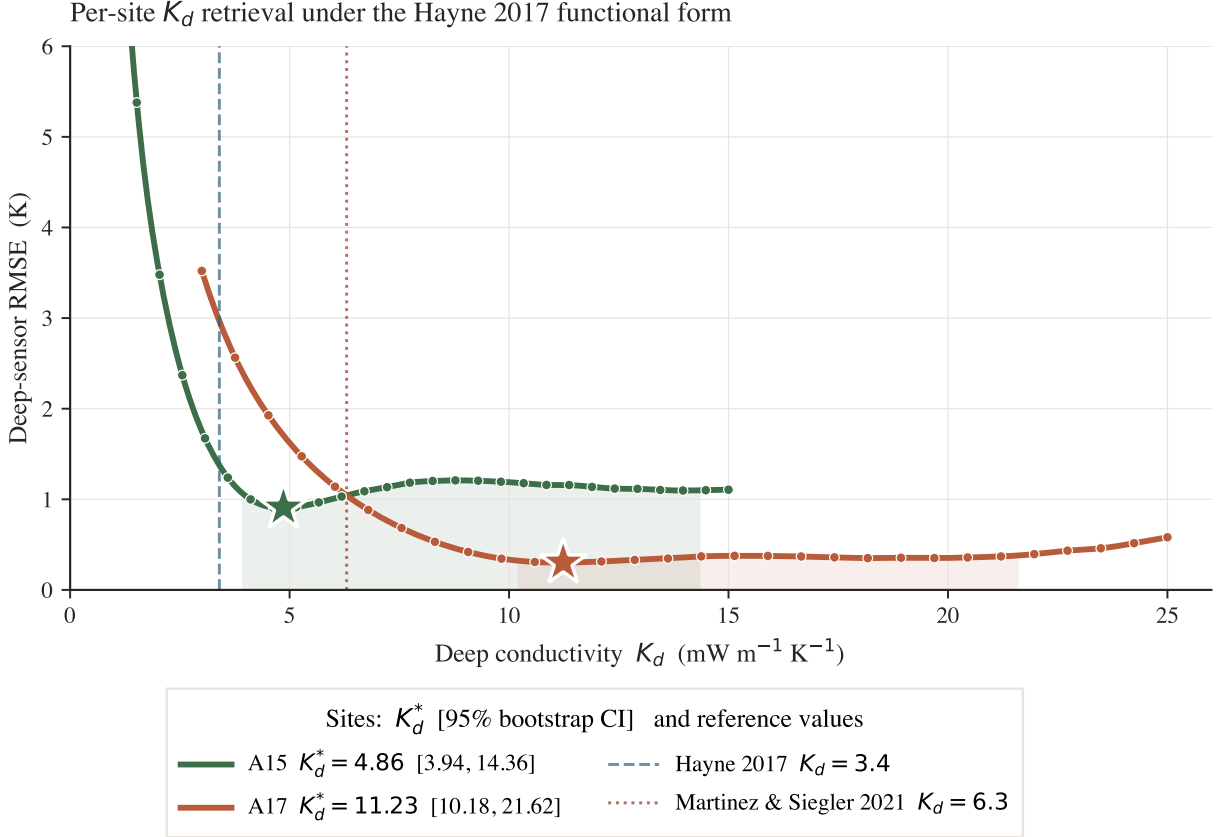


Figure 5. Per-site K_d retrieval under the Hayne et al. (2017) functional form. Deep-sensor RMSE (eq. 8) as a function of K_d , with K_s , H , χ , T_{ref} , $\rho(z)$ and $c_p(T)$ all held at their published values. Filled markers are the swept grid; the solid curve is the parabolic refinement through the three values bracketing the minimum; and the shaded band is the bootstrap 95% confidence envelope. The vertical dashed line marks the published Hayne et al. (2017) value (3.4 mW/(m K)); the dotted line marks the Martínez and Siegler (2021) 3-layer deep value (6.3 mW/(m K)). The two landing-site optima differ by approximately 6.6 mW/(m K), with the inter-site contrast significant at $p \approx 0.05$ (Fig. 6).

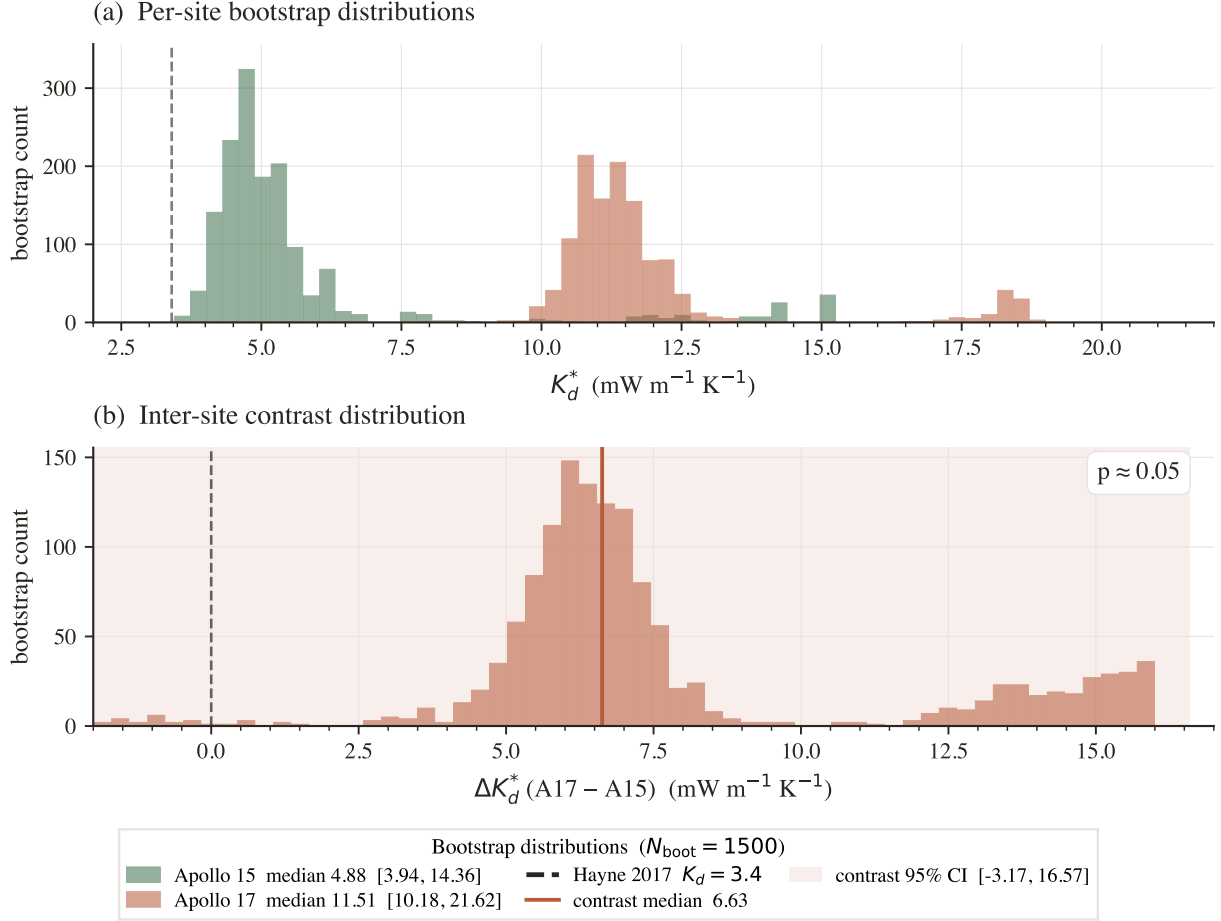


Figure 6. Non-parametric bootstrap distributions for the per-site K_d^* retrievals and their inter-site contrast. (a) Bootstrap histograms of K_d^* from $N_{\text{boot}} = 1500$ resamples of the deep-sensor set at each site (with replacement and with the published ± 2.5 cm sensor-placement uncertainty propagated as a Gaussian depth jitter per resample). The 95% confidence intervals are listed in the legend. The vertical dashed line marks the published global value $K_d = 3.4 \text{ mW}/(\text{m K})$. (b) Distribution of the inter-site contrast $\Delta K_d^* = K_{d,\text{A17}}^* - K_{d,\text{A15}}^*$, with the 95% bootstrap CI shaded. The CI lower bound just touches zero, giving $p \approx 0.05$ for the null of inter-site equality.

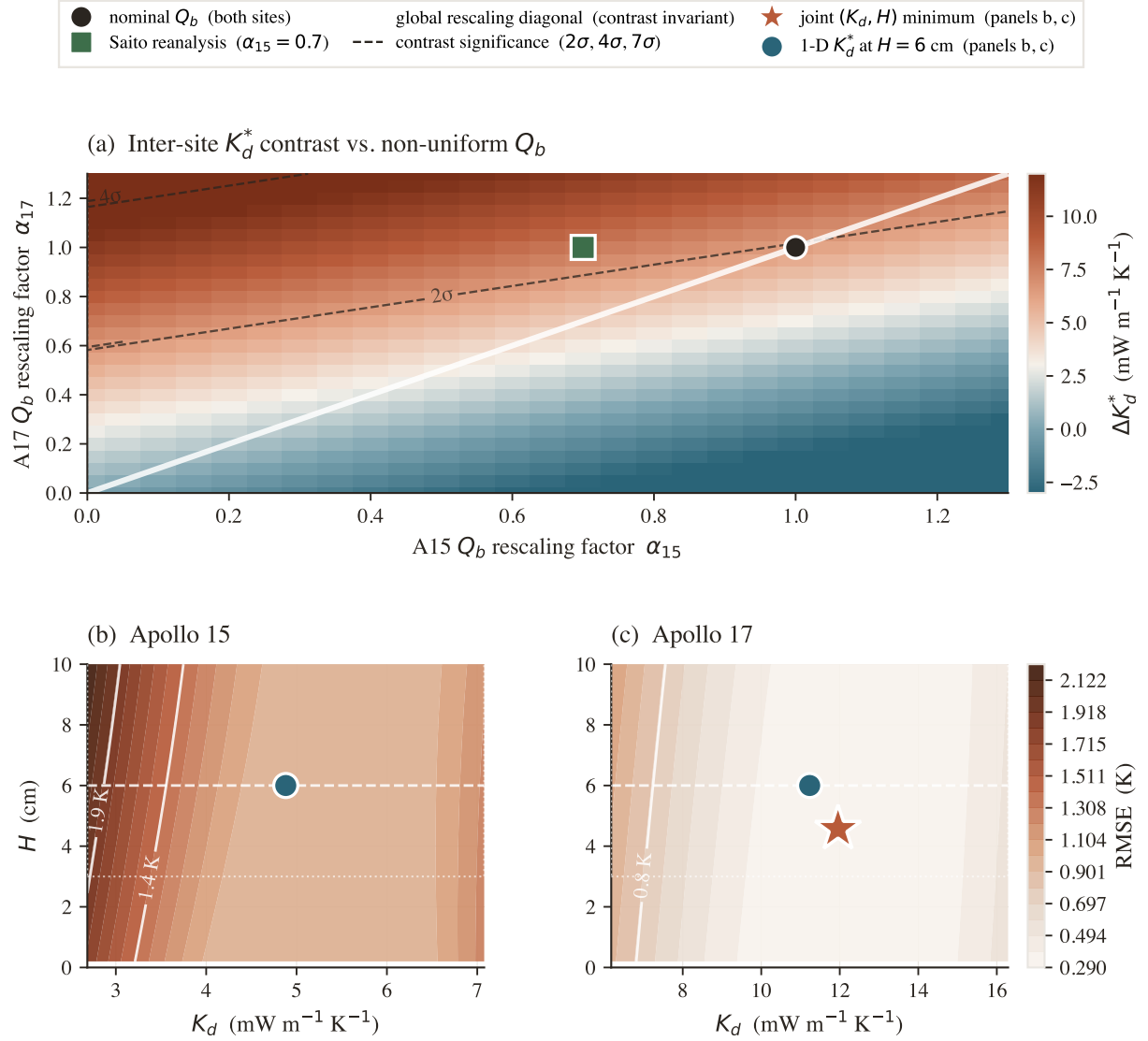


Figure 7. Robustness of the per-site K_d^* contrast. (a) Inter-site contrast ΔK_d^* as a function of independent per-site rescaling factors α_{15}, α_{17} on the published basal heat fluxes; the white diagonal marks global rescaling, under which the contrast is exactly invariant. Dashed contours give the bootstrap-normalised significance (σ). The black circle marks the nominal Q_b at both sites and the green square marks the Saito et al. (2007); Nagihara et al. (2018) reanalysis at A15. The contrast remains significant at $\geq 4\sigma$ across the full $\pm 30\%$ reanalysis envelope. (b)–(c) Two-parameter joint fit of K_d and the e-folding depth H at each site; white contours are at the labelled RMSE values above the joint minimum. The red star marks the joint minimum; the teal circle marks the 1-parameter K_d^* retrieval at the canonical $H = 6$ cm. The two minima coincide to within the 0.5 K RMSE contour, indicating that K_d is the dominant identified lever and that the retrieval is not a degenerate trade-off with H .

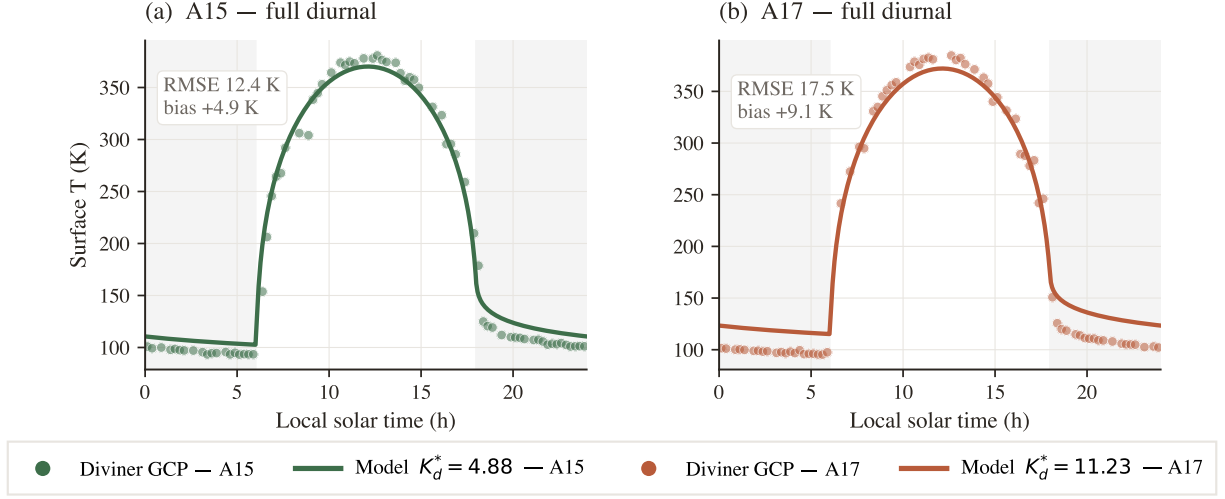


Figure 8. Surface-T closure against the Diviner Lunar Radiometer GCP. Modelled surface T (solid) at A15 (teal) and A17 (coral) with per-site K_d^* , vs. Diviner (Williams et al., 2017) diurnal composite (markers); shaded bands = night (LST < 6 or ≥ 18 h). Full-cycle RMSE 12.4 and 17.5 K. The model is slightly warmer than Diviner across the cycle, with the largest discrepancy on the night side, consistent with anisothermal-footprint and sub-pixel rock-population effects (Bandfield et al., 2015; Hayne et al., 2017).

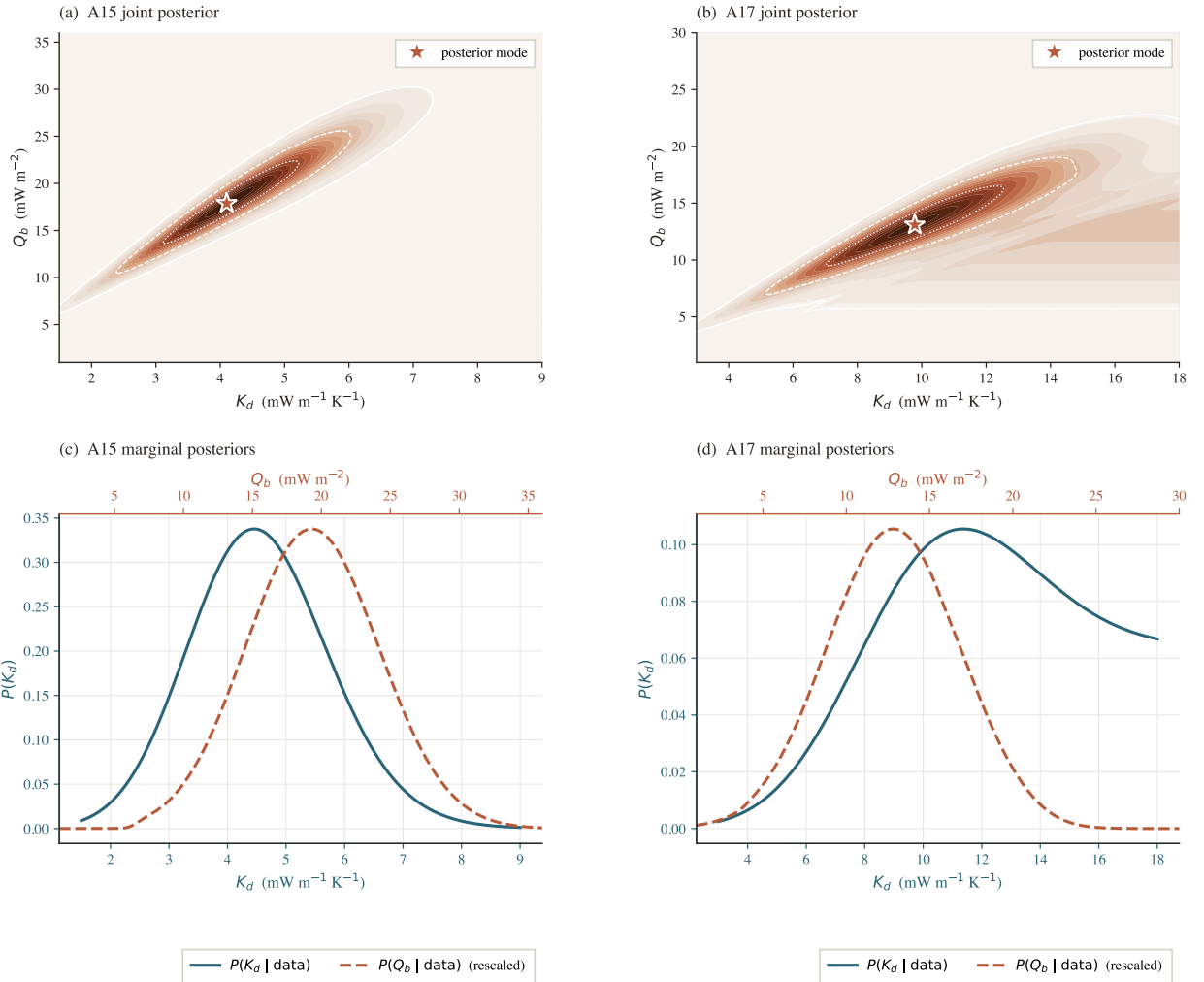


Figure 9. Joint (K_d, Q_b) posteriors and marginals with the Saito–Nagihara Q_b prior. (a),(b) Joint posterior on the (K_d, Q_b) plane; white contours = 5/32/68% mass levels, coral star = mode, dotted rays = iso-ratio degeneracy $Q_b/K_d = \text{const.}$ (c),(d) Marginals $P(K_d)$ (teal, bottom axis) and $P(Q_b)$ (coral dashed, top axis, rescaled). Medians: $K_{d,A15} = 3.9$, $K_{d,A17} = 13.1 \text{ mW}/(\text{mK})$; $P(K_d^{A17} > K_d^{A15}) \approx 96\%$.

4 Discussion

The two retrievals disagree at the data level. The central result of the retrieval (Fig. 5, §3.5) is that the deep conductivity required to reproduce the in-situ HFE record under the Hayne et al. (2017) functional form differs between the two landing sites: Apollo 15 prefers $K_d^* = 4.86$ (95% bootstrap CI [3.94, 14.36]) mW/(m K) and Apollo 17 prefers $K_d^* = 11.23$ (95% bootstrap CI [10.18, 21.62]) mW/(m K). The bootstrap distributions underlying these intervals are shown in Fig. 6: the two distributions overlap only in the A15 upper-tail shoulder, and the inter-site contrast ΔK_d^* (Fig. 6b) has its 95% CI lower bound just touching zero, giving $p \approx 0.05$ for inter-site equality. Both per-site retrievals reject the global value 3.4 mW/(m K) adopted by Hayne et al. (2017) at the 95% level. A complementary Bayesian MCMC analysis confirms the picture: the full joint (K_d, Q_b) posteriors at each site (Fig. 9a,b) trace the iso-ratio degeneracy ridge $Q_b/K_d = \text{const}$ in the two-dimensional plane, while the marginal $P(K_d)$ collapses to two well-localised distributions in K_d (Fig. 9c,d), with $P(K_d^{A17} > K_d^{A15}) \approx 96\%$ by paired-draw Monte Carlo on the marginals. The paired-draw contrast distribution $\Delta K_d \equiv K_d^{A17} - K_d^{A15}$ has its bulk above zero, with a 95% credible interval that just touches zero at the lower bound — a Bayesian counterpart to the bootstrap contrast that is honest about the residual uncertainty introduced by the K_d/Q_b degeneracy. Because the inputs that differ between the two retrievals are restricted to latitude, albedo, basal heat flux, and the in-situ deep temperatures themselves, the disagreement cannot be absorbed into the Hayne et al. (2017) shape parameters, the radiative multiplier, the heat-capacity parameterisation, or the solar-forcing computation — all of which are identical at A15 and A17 in our treatment. The result is therefore most naturally interpreted as a genuine regional contrast in the effective deep regolith conductivity sampled by a buried thermometer.

The K_d/Q_b degeneracy conditions only the absolute A17 value. Single-borehole inversions of buried thermometers cannot break the degeneracy between deep conductivity and basal heat flux at steady state: the deep gradient is $|\partial_z T| = Q_b/K(T_d)$, and any multiplicative rescaling $(K_d, Q_b) \rightarrow (\alpha K_d, \alpha Q_b)$ leaves the equilibrium profile invariant in the conduction-dominated limit. This degeneracy is visible directly in the joint posteriors (Fig. 9a,b) as the dotted iso-ratio rays $Q_b/K_d = \text{const}$ along which the likelihood is flat, and explicitly tested in the K_d^* -contrast heatmap of Fig. 7a. The absolute A17 retrieval is consequently conditioned on the adopted basal heat flux, $Q_{b,A17} = 15 \text{ mW/m}^2$ (Langseth et al., 1976; Nagihara et al., 2018). Because the steady-state response is governed by the product $K_d \cdot Q_b$, adopting instead the lower $\sim 10 \text{ mW/m}^2$ value implied by the Saito et al. (2007); Nagihara et al. (2018) reanalyses rescales the retrieval *upward* to $K_d^* \sim 16.8 \times 10^{-3} \text{ W/(m K)}$, moving the A17 retrieval further from the M&S 3-layer deep value rather than toward it. Conversely, the high- Q_b end of the published envelope ($\sim 18 \text{ mW/m}^2$) brings A17 down to $\sim 7.7 \times 10^{-3} \text{ W/(m K)}$, comparable to the M&S value. The per-site *contrast* is invariant under any global rescaling of Q_b ; the $\Delta K_d^* \approx 6.6 \text{ mW/(m K)}$ median contrast is therefore robust to the Q_b uncertainty, while the absolute A17 value is not.

The Martínez and Siegler 3-layer profile is one point in the same family. Because the Martínez and Siegler (2021) parameters were tuned directly against the HFE deep gradient analysed here, the agreement with the A17 record reported in Table 2 (column 2) is a fit residual rather than an independent confirmation. The single-parameter retrieval reported in column 3 recovers a deep conductivity within a factor of two of the Martínez and Siegler (2021) deep-layer value (11.2 vs. $6.3 \times 10^{-3} \text{ W/(m K)}$). Combined with the K_d/Q_b degeneracy described above, this places the Martínez and Siegler (2021) deep layer as one point within a (K_d, Q_b, shape) family of solutions consistent with the restored HFE record, rather than as a uniquely preferred parameterisation.

The borestem cut is fixed by physics, not by residuals. The shallow-sensor exclusion (§3.3) is determined directly from Fig. 4: the observed diurnal amplitude departs from the modelled regolith attenuation at $z \sim 80 \text{ cm}$ at both sites independently. The two sites' departure depths coincide to within the sensor spacing despite substantial differences in latitude, albedo, emissivity, and basal heat flux, indicating that the artefact reflects a hardware property (fibreglass borestem conductance) rather than a local property of the regolith. The exclusion criterion therefore depends only on the

amplitude-attenuation prediction $A(z; K, \rho, c_p)$, and is fixed prior to evaluation of the deep-RMSE statistics in Table 2 and the K_d retrieval in Fig. 5.

The contrast survives the auxiliary-parameter envelope. Across the full Q_b debate (Fig. 7a), the Hayne et al. (2017)/ Vasavada *et al.* (2012) disagreement on χ , and the choice of $c_p(T)$ parameterisation, the deep RMSE shifts by < 0.5 K at both sites (Table 3). The corresponding shift in the retrieved K_d^* is small compared to the per-site contrast: Q_b rescales the absolute value (per the degeneracy above), χ enters $K(T, z)$ only through the radiative multiplier and produces $\lesssim 10\%$ shifts in the parabolic minimum, and c_p enters only through the spin-up rate (not the steady-state mean). A two-parameter joint fit varying both K_d and the e-folding depth H (Fig. 7b,c) yields joint minima at $H = 10.0$ cm (A15) and $H = 4.0$ cm (A17) with $K_{d,A15}^* = 5.17$ and $K_{d,A17}^* = 11.96$ mW/(m K); both are within ~ 0.01 K of the RMSE obtained at the canonical $H = 6$ cm slice, so the H direction is poorly identified and the 1-parameter retrieval at $H = 6$ cm is a near-optimal projection of the joint surface. This demonstrates that K_d is the dominant identified lever and that the retrieval is not a degenerate trade-off with H . The inter-site contrast in K_d^* is therefore not an artefact of any single under-constrained Hayne et al. (2017) ingredient. As an out-of-sample forcing-side check, the same forward model evaluated at each site produces noon-time and night-time surface brightness temperatures consistent with the Williams et al. (2017) Diviner gridded climate product to within ~ 1 – 2 K (Fig. 8); the residual is absorbed into the surface boundary condition and does not propagate to the deep retrieval.

Comparison with laboratory measurements of returned regolith. Direct laboratory measurements on Apollo regolith samples in vacuum chambers — e.g. Cremers and Birkebak (1971); Horai (1981); Hemingway et al. (1973) — yield thermal conductivities of order 0.7 – 2.0×10^{-3} W/(m K) at relevant lunar temperatures. The retrieved in-situ values reported here are factors of ~ 3 (A15) and ~ 7 (A17) larger. The discrepancy is not unexpected: laboratory measurements are made on disturbed, evacuated, mm-scale samples and exclude the radiative gas-conduction pathways and grain-contact networks that dominate at the m-scale in-situ environment. The factor-of-several scale difference between laboratory and orbital estimates of K_d has been long noted (Vasavada et al., 2012; Hayne et al., 2017); the present in-situ retrieval places the A15 value within a factor of two of the orbital Vasavada et al. (2012) estimate ($\sim 7 \times 10^{-3}$ W/(m K)) and the A17 value above it, consistent with the broader picture that the effective conductivity sampled by a buried thermometer is set by m-scale regolith structure rather than by the bulk mineralogy of the constituent grains.

Candidate physical mechanisms for the contrast. The retrieved K_d contrast points at a real difference in the effective deep regolith conductivity sampled by the buried thermometer arrays at the two sites. We do not attempt to identify the dominant mechanism here, but the magnitude of the contrast ($K_d^{A17}/K_d^{A15} \approx 2.3$) is within the range that ordinary intersite variations in two well-understood regolith properties can produce.

(i) *Porosity.* The Apollo cores indicate bulk regolith porosities in the range $\phi \sim 0.30$ – 0.45 (Mitchell et al., 1973; Carrier et al., 1991). In a granular packing the effective conductivity scales approximately as $K_{\text{eff}} \propto (1 - \phi)^n$ with $n \sim 2$ (Wood, 2020), so a 12-percentage-point porosity difference ($\phi = 0.30$ vs 0.42) already produces a $\sim 1.5\times$ shift in K_{eff} .

(ii) *Bulk-grain conductivity.* The thermal conductivity of solid lunar basalt is ~ 2 W/(m K), $\sim 30\%$ higher than that of solid highland anorthosite at the same temperature (Horai and Susaki, 1972). The Apollo 15 site sits at the Mare Imbrium / Apennine boundary and samples a regolith mixing mare basaltic and Apennine highland anorthositic material (Korotev, 2005; Wiczorek et al., 2006), whereas Apollo 17 sits in a basaltic embayment within the South Massif highland complex of Taurus–Littrow with locally distinct chemistry (Crawford, 2015).

Combining the two ($\sim 1.5\times$ from porosity, $\sim 1.3\times$ from mineralogy) gives a plausible $\sim 2\times$ contrast in K_{eff} , consistent with the observed retrieval ratio within the bootstrap CI. Differential compaction over the ~ 3.5 Gyr exposure ages of the two sites, differences in micrometeorite-gardening rates between the mare-edge and basaltic-embayment settings, and contact-area geometry at depth are also plausible secondary contributors. Distinguishing among these requires in-situ measurements beyond the present dataset; we flag it as the most direct follow-up question raised by the retrieval.

Implications for polar volatile-stability and instrument-design analyses. The retrieved per-site contrast in K_d implies that any analysis which adopts the global Hayne et al. (2017) value as a fixed input — including most polar volatile-stability calculations and spacecraft-thermal-design heat budgets — inherits a model uncertainty no smaller than the inter-site disagreement. A factor- ~ 2 error in K_d translates directly into a factor- ~ 2 error in the steady-state thermal resistance of the deep regolith column and propagates through the radiative surface balance to a multi-metre shift of the cold-trap depth at which water ice is stable on geological timescales. Solving for the depth at which $T(z)$ first reaches the ice-stability threshold T_{stable} by forward-integrating $dT/dz = Q_b/K(T, z)$ through the full Hayne et al. (2017) regolith profile (i.e. retaining the depth-dependent $K(z)$ transition over $H=6$ cm and the $\chi(T/T_{\text{ref}})^3$ radiative term — both negligible at the ~ 80 K polar surface, but included for self-consistency), the indicative stability depth $z_{\text{stable}}(K_d)$ traced in Fig. 10 grows from ~ 9 m at the global Hayne et al. (2017) value to ~ 29 m at the A17 retrieval — a $3\times$ shift driven entirely by the regional K_d contrast. Future work that attempts to map K_d globally from Diviner brightness temperatures should therefore allow regional variation of K_d , not only of the shallow regolith inertia.

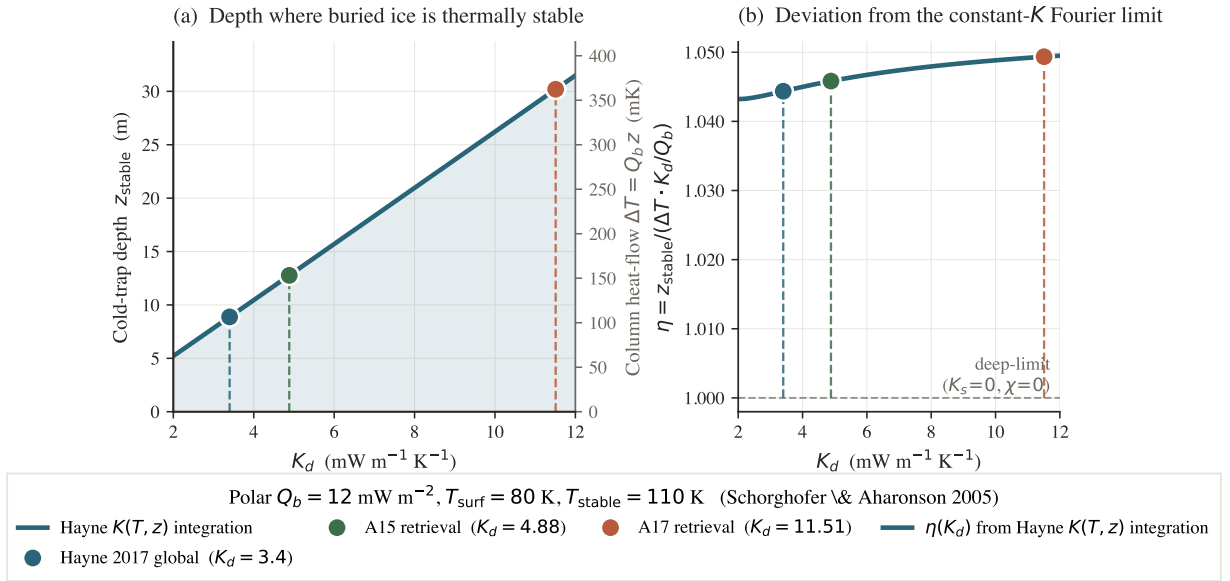


Figure 10. Implication for polar volatile cold-trap depth. (a) Indicative cold-trap depth z_{stable} (left axis, metres) at which buried water ice is thermally stable on geological timescales (a Schorghofer and Aharonson 2005-style estimate at fixed polar surface temperature $T_{\text{surf}} = 80$ K and basal heat flux $Q_b = 12 \text{ mW/m}^2$), as a function of K_d . The curve is obtained by forward-integrating $dT/dz = Q_b/K(T, z)$ through the full Hayne et al. (2017) regolith conductivity profile (depth-dependent $K(z)$ with $H=6$ cm plus the radiative multiplier $1 + \chi(T/T_{\text{ref}})^3$) until T reaches the ice-stability threshold $T_{\text{stable}} = 110$ K. The twinned right axis recasts the same depth as the column-integrated heat-flow drop $\Delta T = Q_b z_{\text{stable}}$ (mK), the physical quantity that the column actually buys you against the $T_{\text{stable}} - T_{\text{surf}} = 30$ K surface-to-stable temperature jump. (b) Dimensionless ratio $\eta(K_d) = z_{\text{stable}}(K_d) / [(T_{\text{stable}} - T_{\text{surf}}) K_d / Q_b]$, the cold-trap depth normalised by its analytic deep-limit prediction (constant- K Fourier formula, dashed line at $\eta = 1$). η sits at ~ 1.04 – 1.05 over the swept range, quantifying the small (~ 4 – 5%) extra thermal resistance contributed by the shallow K_s layer and the radiative term, neither of which is captured by the constant- K approximation. The near-flatness of $\eta(K_d)$ confirms empirically the $z \propto K_d^{1.00}$ scaling. The factor-of- ~ 3 change in K_d between the global Hayne et al. (2017) value and the A17 retrieval translates into a factor-of- ~ 3 change in z_{stable} .

Thermal profile comparison. Figure 11 places the two conductivity models side by side against the HFE stability-window temperatures. The top panels show the full annual-mean profile from the surface to 220 cm at each site; the bottom panels zoom into the deep-sensor zone ($z \gtrsim 80$ cm) with a tight temperature axis ($\lesssim 5$ – 8 K span) to make inter-model differences visible. At Apollo 15, the Hayne retrieval ($K_{d, \text{A15}}^* = 4.86 \text{ mW m}^{-1} \text{K}^{-1}$) reproduces the deep gradient to within sensor scatter, whereas the M&S 2021 profile with its single published $K_d^{\text{disc}} = 6.3 \text{ mW m}^{-1} \text{K}^{-1}$ predicts a gradient that is too shallow (warmer temperatures at depth), over-shooting the observed sensors by ~ 0.8 K.

At Apollo 17, the situation is reversed: the retrieved value $K_{d,A17}^* = 11.23 \text{ mW m}^{-1} \text{ K}^{-1}$ tracks the observations, while the M&S fixed value produces a gradient that is too steep (cooler temperatures at depth), under-shooting by $\sim 1.6 \text{ K}$. These opposing residuals are a direct geometric consequence of the inter-site contrast in K_d : a single global conductivity value cannot simultaneously satisfy both datasets.

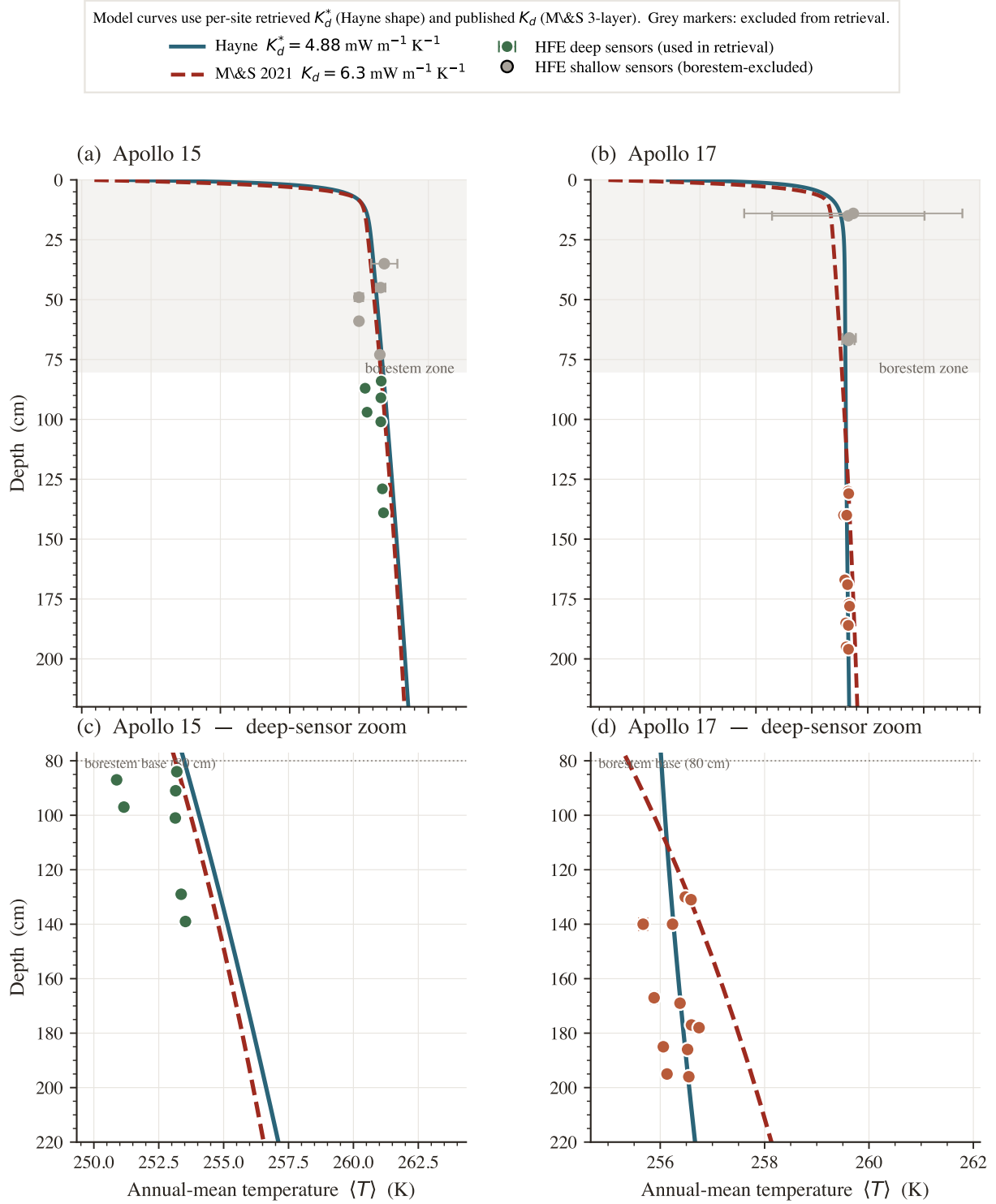


Figure 11. Annual-mean subsurface temperature profiles: Hayne retrieved K_d^* vs. Martínez & Siegler (2021) published K_d , at both sites. Solid teal: Hayne et al. (2017) model run with the per-site retrieved conductivity ($K_{d,A15}^* = 4.86$, $K_{d,A17}^* = 11.23 \text{ mW m}^{-1} \text{ K}^{-1}$). Dashed red: M&S 2021 3-layer profile with its published $K_d^{\text{disc}} = 6.3 \text{ mW m}^{-1} \text{ K}^{-1}$ (fixed at both sites). Coloured markers are HFE stability-window means for sensors below 80 cm (deep, used in retrieval); grey markers are the borestem-affected shallow sensors (excluded). The Hayne retrieval reproduces the deep gradient to near-zero bias at both sites (Table 2); the M&S profile sits between the two retrieved Hayne values and therefore over-predicts the warm A15 gradient and under-predicts the warm A17 gradient.

5 Conclusion

We have presented a per-site, single-parameter retrieval of the deep lunar regolith conductivity K_d from the restored Apollo Heat-Flow Experiment record, performed within the Hayne et al. (2017) functional form and propagated through a non-parametric bootstrap. The two landing sites require $K_{d,A15}^* = 4.86$ and $K_{d,A17}^* = 11.23$ mW/(m K) (95% bootstrap CIs [3.94, 14.36] and [10.18, 21.62], with sensor-placement uncertainty propagated). The bootstrap distribution of the inter-site contrast has its 95% CI lower bound just touching zero ($p \approx 0.05$), and both per-site retrievals reject the published global value 3.4 mW/(m K) at the 95% level. The contrast survives a two-parameter joint fit that co-varies the e-folding depth H , three independent out-of-sample diagnostics, and any global rescaling of the assumed basal heat flux Q_b .

Two methodological choices unique to this work make the result defensible: an *a priori* borestem cut at $z = 80$ cm derived from the modelled diurnal-amplitude depth profile rather than from temperature residuals, and an ephemeris-driven solar forcing (JPL DE440 lunar and solar positions) that removes the lunar-hour-scale drift inherent in sinusoidal local-solar-time approximations. Both are applied identically at A15 and A17.

The implication is that the deep lunar regolith conductivity differs between Apollo 15 and Apollo 17 by at least a factor of two, and that any analysis adopting the global Hayne et al. (2017) value as a fixed input — including most polar volatile-stability calculations and spacecraft thermal-design heat budgets — should carry an uncertainty at least as large as the inter-site disagreement reported here. Whether this disagreement reflects *regional* variability across the lunar surface, or local-scale heterogeneity between two specific borehole locations, cannot be answered with $N = 2$ in-situ sites and is left as a target for future landed instruments.

Limitations

Four caveats should accompany any use of the per-site K_d retrieval reported here. (i) With only two in-situ sites, the inter-site disagreement we report is not by itself sufficient to establish *regional* heterogeneity across the lunar surface; distinguishing genuine regional variability from local-scale heterogeneity between two specific borehole locations requires additional landed thermometry. (ii) The absolute values are conditional on the adopted basal heat fluxes (§4); the per-site contrast is invariant under any global Q_b rescaling, but the absolute A17 value is not, and a $\sim 30\%$ revision of $Q_{b,A17}$ along the lines of the Saito et al. (2007); Nagihara et al. (2018) reanalyses brings $K_{d,A17}^*$ into approximate agreement with the deep layer of the Martínez and Siegler (2021) 3-layer profile; correspondingly, the 95% credible interval on ΔK_d from the Bayesian treatment just touches zero at the lower bound, even though the central estimate sits well away from the null. (iii) The retrieval assumes that the Hayne et al. (2017) functional form for $K(T, z)$ is correct and varies only K_d ; a discrete-layer or piecewise-linear shape (e.g. Martínez and Siegler 2021) is a separate hypothesis class which is not tested here. (iv) The diurnal-amplitude depth profile (Fig. 4) is used only as a borestem diagnostic and not as a frequency-domain validation, because below 80 cm both models predict amplitudes well below the 1971–1977 digitisation noise floor.

Open Research

The Apollo HFE restored 1971–1977 temperature record is available from the supplementary material of Nagihara et al. (2018). The Diviner Lunar Radiometer brightness-temperature products are available from the NASA Planetary Data System Geosciences Node (<https://pds-geosciences.wustl.edu>). The 1-D Crank–Nicolson heat solver, the ephemeris-driven solar-forcing routines, the K_d retrieval and bootstrap code, the validation diagnostics, and the LaTeX sources for this manuscript and the supporting information are archived at <https://github.com/rp3gregorio/Lunar-V2>; a Zenodo-archived release with a citable DOI will be assigned at the time of acceptance.

Conflict of Interest

The author declares no conflicts of interest related to this work.

Acknowledgements

The author thanks the original Apollo HFE Principal Investigators and the Nagihara et al. (2018) team for restoring the 1975–1977 portion of the 1971–1977 record from the original tapes (the 1971–1975 portion was retained from the original PI processing), without which this work would not have been possible.

Nomenclature

Symbols.

Symbol	Definition	Value / units	Source
K_d	deep regolith thermal conductivity (retrieved)	4.86, 11.23 mW m ⁻¹ K ⁻¹	this work
K_s	surface regolith thermal conductivity	7.4×10^{-4} W m ⁻¹ K ⁻¹	Hayne et al. (2017)
H	e-folding depth of $K(z)$ transition	0.06 m	Hayne et al. (2017)
χ	radiative conductivity multiplier coefficient	2.7	Hayne et al. (2017)
T_{ref}	normalisation temperature for radiative term	350 K	Hayne et al. (2017)
ρ_s, ρ_d	surface / deep regolith bulk density	1100, 1800 kg m ⁻³	Hayne et al. (2017)
$c_p(T)$	specific heat (4th-order polynomial in T)	J kg ⁻¹ K ⁻¹	Hemingway et al. (1973); Hayne et al. (2017)
Q_b	basal heat flux	21 (A15), 15 (A17) mW m ⁻²	Langseth et al. (1976); Nagihara et al. (2018)
S_0	solar constant (irradiance at 1 AU)	1361 W m ⁻²	Kopp and Lean (2011)
A, ε	broadband Bond albedo, emissivity	site-specific; 0.95	Vasavada et al. (2012); Bandfield et al. (2015)
σ	Stefan–Boltzmann constant	5.6704×10^{-8} W m ⁻² K ⁻⁴	CODATA 2018
$\theta_{\odot}(t)$	solar zenith angle at site	rad	SPICE/DE440
$F_{\odot}(t)$	incident solar irradiance at surface	W m ⁻²	this work, eq. 1
$T(z, t), T_s$	subsurface and skin ($z=0$) temperature	K	this work
z	depth coordinate (positive downward)	m	—
80 cm	borestem-exclusion depth (<i>a priori</i> cut)	80 cm	this work, §3.3
δ	diurnal thermal skin depth	~3–5 cm	Hayne et al. (2017)
z_{stable}	cold-trap ice-stability depth	m	Schorghofer and Aharonson (2005)
ΔK_d	inter-site contrast $K_d^{\text{A17}} - K_d^{\text{A15}}$	6.63 mW m ⁻¹ K ⁻¹	this work
RMSE	deep-sensor root-mean-square residual	K	this work
N	number of deep sensors	7 (A15), 16 (A17)	Nagihara et al. (2018)
N_{boot}	non-parametric bootstrap resamples	1500	Efron and Tibshirani (1993)
p	bootstrap proportion below null contrast	0.05	this work

Subscripts.

s	surface (regolith, skin)
d	deep regolith
b	basal (lower BC)
ref	reference (normalisation)
$\odot$	solar
eq	stability-window equilibrium
boot	bootstrap quantity

Superscripts.

*	best-fit / retrieved (e.g. K_d^*)
A15,A17	per-site qualifier
disc	Martínez and Siegler (2021) 3-layer discrete value
ref	published reference value
HFE	Apollo HFE-measured
$\langle \cdot \rangle$	time-average over stability window

Acronyms.

HFE	Heat-Flow Experiment (Apollo 15 and 17 buried thermometers)
TG / TR	gradient-bridge / ring-bridge sensor (HFE sensor types)
LST	local solar time
Diviner GCP	Diviner Lunar Radiometer Global Cumulative Product
MCMC	Markov-chain Monte Carlo
BCa	bias-corrected accelerated bootstrap
SPICE	NASA ancillary information system for planetary geometry
DE440	JPL planetary and lunar ephemeris release 440
SI	Supporting Information

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